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1. Administrative & Study Identification

Full Study Title: Study of 2 Medicines (Aztreonam and Avibactam) Compared to Best Available Therapy for Serious Gram-negative Infections **Short Title/Acronym:** ATM-AVI Pediatric Phase 2A Study **Protocol Number:** C3601008 **NCT Number:** NCT05639647 **Sponsor Name:** Pfizer **Investigational Product:** Aztreonam-Avibactam (ATM-AVI) $\pm$ Metronidazole (MTZ) **Phase of Development:** Phase 2A **Trial Status:** Recruiting **Medical Monitor:** Pfizer Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the pharmacokinetics (PK), safety, and tolerability of Aztreonam (ATM) and Avibactam (AVI) in pediatric participants with serious gram-negative bacterial infections. * **Secondary Objectives:** To assess the clinical efficacy, measured by favorable clinical response (CR) at the End of IV Treatment (EOIV), End of Treatment (EOT), and Test of Cure (TOC), of ATM-AVI compared to best available therapy (BAT).

2.2 Background & Rationale To address the critical medical need for effective treatments against complicated and resistant gram-negative bacterial infections in pediatric patients. Complicated intra-abdominal infections (cIAI), complicated urinary tract infections (cUTI), hospital-acquired/ventilator-associated pneumonia (HAP/VAP), and bloodstream infections (BSI) are frequently caused by pathogens resistant to standard antibiotics. The ATM-AVI combination specifically targets these resistant strains, including metallo-β-lactamase-producing organisms, preventing bacteria from resisting antibiotic treatment.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Best Available Therapy [BAT]) * **Blinding:** Open-label, utilizing a single-blinded observer assigned by each investigational site to assess clinical outcomes. * **Randomization:** 3:1 ratio (ATM-AVI : BAT).

3.2 Planned Enrollment & Duration * **Target N:** 48 participants (divided into 4 age cohorts of 12 participants each: Cohort 1 [12 to <18 years], Cohort 2 [6 to <12 years], Cohort 3 [2 to <6 years], Cohort 4 [9 months to <2 years]). If the ATM-AVI dose requires adjustment during reviews, Cohort 4 will be increased to

20 participants (15 receiving ATM-AVI, 5 BAT). * **Number of Sites:** Approximately 33 international, multicenter sites. * **Estimated Study Start:** 12-Dec-2022 * **Estimated Primary Completion:** 12-Dec-2025

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 9 months to < 18 years at screening. 2. Suspected or confirmed cIAI, cUTI, HAP/VAP, or BSI with gram-negative pathogens. 3. Requires hospitalization and intravenous (IV) antibiotic treatment. 4. Female (post-menarchal) participants must have a negative serum/urine pregnancy test (β hCG sensitivity ≥ 25 mIU/mL).

4.2 Exclusion Criteria 1. Pregnant or breastfeeding; fertile male/female unwilling/unable to use effective contraception for ≥ 7 days (males) or ≥ 28 days (females) after the last ATM-AVI infusion. 2. Estimated Creatinine Clearance (CrCL) ≤ 15 mL/min/1.73 m2 (eCrCl or eGFR calculation based on age). 3. Non-infectious related screening ALT or AST $>3 \times$ ULN, ALP $>3 \times$ ULN, and/or Total Bilirubin $>2 \times$ ULN ($>3 \times$ ULN for Gilbert's syndrome). 4. Current use of prohibited concomitant medication(s), unwilling/unable to use Metronidazole (MTZ) for cIAI (Cockayne Syndrome patients with cIAI are excluded due to hepatotoxicity risks), or having received previous investigational drug(s) ≤ 30 days or 5 half-lives before randomization. 5. Received >24 hours of systemic antibiotics during the 48 hours before randomization unless participant documented treatment failure after at least 48 hours of antibiotic therapy. 6. Gram-negative species not expected to respond to ATM-AVI ≤ 14 days. 7. **(HAP/VAP only):** Microbiologically known or high likelihood of monomicrobial infection with a gram-positive organism, lung abscess, pleural empyema, post-obstructive pneumonia, or lung/heart transplant.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Initial ATM-AVI IV loading dose, followed by maintenance doses every 6 or 8 hours depending on participant weight and kidney function. Participants with cIAI (and others at investigator discretion for anaerobic infections) will also receive IV Metronidazole (MTZ). * **Route of Administration:** Intravenous (IV) infusion given over 3 hours (MTZ infused over 20-30 minutes). Switch to oral therapy is allowed starting on Day 4 if clinical improvement is observed. * **Duration of Treatment:** Up to a maximum of 14 days of ATM-AVI treatment.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Systemic IV antivirals, non-antibiotic lavage. Optional gram-positive coverage is permitted (vancomycin, teicoplanin, and linezolid; daptomycin is only permitted for cIAI and BSI). IV aminoglycoside coverage is permitted for *Pseudomonas aeruginosa*. * **Prohibited:** Meropenem/carbapenem-type antibiotics, probenecid, inhaled antibiotics, antibiotic peritoneal lavage. Systemic antibiotics are not permitted unless the study drug failed or for a remote site infection. Systemic antifungals should be avoided unless strictly indicated. Participants with cUTI may not take prophylactic therapy during study drug treatment.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -2 to 0	Informed Consent, eligibility assessment (must be within 48h of randomization)
Baseline	Day 1	Randomization (3:1), First IV Dose, baseline PK blood draws
Interim	Days 1 - 14	IV Treatment, 5 PK micro-blood draws during first 2 days. Day 4: assessment for oral switch
End of Study	Days 33 - 50	Test of Cure (TOC) assessment 7-14 days post-treatment; phone follow-up 28-35 days post-treatment

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Definition:** Pharmacokinetics (PK) and Safety. Measurement of maximum predicted plasma concentration ($Cmax$), minimum predicted trough plasma concentration ($Cmin$), area under the concentration-time curve (AUC), Clearance, and Half-life of ATM and AVI. * **Measurement Method:** 5 PK micro-blood samples of at least 0.1 mL drawn (first 2 samples at 2h and 5h after first infusion; next 3 samples around the 5th or later infusion) and evaluated via population pharmacokinetic (popPK) analysis.

7.2 Secondary & Exploratory Endpoints * **Clinical Efficacy:** Percentage of participants with favorable clinical response (CR) evaluated by a blinded observer at the following timepoints: 1. End of IV study treatment (EOIV) 2. End of Treatment (EOT) 3. Test of Cure (TOC) within 7 to 14 days of the last antibiotic treatment. Cure is defined as signs and symptoms improved with no further antimicrobial treatment required.

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard vital and lab monitoring with a specific focus on potential liver function abnormalities and acute kidney injury. The blinded observer will evaluate AE/SAE causality while in-hospital, within 24 hours of EOIV, and 48 hours of EOT for those on oral step-down.
- **Serious Adverse Events (SAEs):** Standard reporting timelines within 24 hours of site awareness.

- **Data Monitoring Committee (DMC):** An independent external DMC and the Sponsor review PK data and safety. For Cohort 3, enrollment begins after the first 4 participants complete Cohort 2. For Cohort 4, enrollment begins after the first 4 participants complete Cohort 3. Enrollment is not paused during the PK data reviews of the first 4 ATM-AVI participants in a cohort.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set, Pharmacokinetic (PK) Analysis Set, and Intent-to-Treat/Efficacy Set.
- **Sample Size Justification:** Target $N=48$ appropriately sized to define PK parameters and ensure safety profiles in pediatric strata prior to broader efficacy trials. If dose adjustment is needed based on DMC review, Cohort 4 will be expanded.
- **Handling of Missing Data:** Handled via predictive Population Pharmacokinetic (popPK) modeling.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Regular onsite and remote monitoring. Implementation of a single-blinded observer assigned by each site specifically to evaluate clinical responses to minimize bias in this open-label design.
- **Audit Trail:** eCRF data entry, central laboratory PK tracking, and data clarification forms.

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11. Study Identifiers & Governance

- **Primary ID:** C3601008
- **Registry Number:** NCT05639647
- **Sponsor/Lead Organization:** Pfizer
- **Study Title:** Study of 2 Medicines (Aztreonam and Avibactam) Compared to Best Available Therapy for Serious Gram-negative Infections
- **Official Regulatory Title:** A PHASE 2A MULTICENTER, OBSERVER-BLINDED, RANDOMIZED 2 ARM STUDY TO INVESTIGATE PHARMACOKINETICS, SAFETY, TOLERABILITY AND EFFICACY OF INTRAVENOUS AZTREONAM-AVIBACTAM $\pm$ METRONIDAZOLE COMPARED TO BEST AVAILABLE THERAPY (BAT) IN PEDIATRIC PARTICIPANTS 9 MONTHS TO LESS THAN 18 YEARS OF AGE WITH SERIOUS GRAM-NEGATIVE BACTERIAL INFECTIONS INCLUDING COMPLICATED INTRA-ABDOMINAL INFECTION

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12. Clinical Framework (Infectious Disease Specific)

- **Primary Condition:** Serious Gram-negative Bacterial Infections (cIAI, cUTI, HAP/VAP, BSI)
- **Diagnosis Threshold:** Suspected/known gram-negative infection requiring hospitalization and IV antibiotics
- **Medical Methodology:** Intravenous (IV) Antibiotic Therapy
- **Optimization Target:** Pathogen eradication and clinical symptom resolution (Cure)

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13. Study Procedures & Methodology

- **Management Pathway:** Best Available Therapy (BAT) versus ATM-AVI protocol
- **Education Protocol (HCP):** Site training on micro-blood PK sampling, single-blind observer masking protocols, and MTZ administration (monitoring for hepatotoxicity/Cockayne syndrome exclusion).
- **Education Protocol (Patient):** Caregiver/Parental informed consent and education on IV-to-oral switch conditions.
- **Quality Audit Mechanism:** Independent Data Monitoring Committee (DMC) safety reviews triggered sequentially after the first 4 participants of designated cohorts complete treatment.

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14. Observation & Follow-Up Schedule

- **Total Duration:** 33 to 50 days
- **Onsite Visit Cadence:** Daily while hospitalized (up to 14 days of IV treatment)
- **Remote Monitoring Frequency:** Telephone contact 28 to 35 days after the last study treatment to check health status
- **Checklist Review Interval:** Test of Cure (TOC) assessment 7 to 14 days following the end of antibiotic therapy

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]